Year 8 Science – Booklet 12: Biology

Microbes & Disease (Fighting Disease) — ANSWERS

Note: this booklet's content (“Fighting disease”) is essentially identical to Year 9 Booklet 6 (Disease & Immunity), so the same worked answers apply throughout.

Quick Test (p.3)

1. Bacteria, viruses, and fungi.
2. Antitoxins and antibodies.
3. A vaccine contains dead or harmless microorganisms (with antigens on them); it trains white blood cells to make antibodies in advance, so the body can respond immediately if it meets the real microorganism later.
4. Any two of: AIDS (caused by HIV), flu, chicken pox, measles.
5. Any two of: food poisoning, chlamydia, whooping cough, tetanus.

Section A – Multiple Choice (5 marks)

1. d) bacteria
2. b) HIV
3. b) viruses
4. c) fungi
5. c) white blood cells

Section B (13 marks)

1. a) Three ways disease-causing bacteria can enter your body (3 marks)

Through the air (breathing in droplets from a cough/sneeze); via contaminated food or water; through cuts or wounds in the skin, or contact with an infected person.

1. b) How white blood cells fight bacteria (2 marks)

Some white blood cells engulf and destroy the bacteria directly; others produce antibodies that attach to antigens on the bacteria's surface and clump them together so they can be engulfed, and antitoxins that neutralise any toxins the bacteria release.

2. Extra help fighting disease

a). Before infection: vaccination (artificial immunity gives the immune system advance warning).

b). After infection: antibiotics (medicines that kill bacteria – note these don't work on viruses).

3. Two other types of microorganism that may cause disease (2 marks)

Viruses and fungi.

4. Unscramble infectious diseases (4 marks)

•. nteaust → **tetanus**

•. sotbeculsuir → **tuberculosis**

•. nhcekic xpo → **chicken pox**

•. leaesms → **measles**

Section C (7 marks)

1. Bacterial cell vs virus

a). Similarity (1 mark): both have genetic material that is not enclosed in a nucleus.

b). Difference (1 mark): a bacterial cell has a cell wall, cell membrane and cytoplasm (a complete, living cell); a virus is much simpler, just genetic material surrounded by a protein coat, and is not a complete cell.

c). How vaccines help against viruses (2 marks): a vaccine contains a dead or harmless form of the virus, which still has the virus's antigens. This trains white blood cells to recognise the antigen and produce antibodies in advance, so if the real virus infects the body later, the immune system can respond immediately and destroy it before illness develops.

2. Bacteria growing at different temperatures

a). Tick **37°C** (human body temperature – the temperature at which bacteria that live in the human body reproduce fastest).

b). Two safety precautions (2 marks): 1. Sterilise equipment before and after use, and work near a Bunsen burner flame to reduce contamination. 2. Seal the petri dish lid (e.g. with tape in a cross pattern, not fully sealed) and never open it once bacteria have grown; wash hands thoroughly afterwards.